

Study Design: Deep Learning for PeptideMHC Binding Prediction

Mimicking & Extending Jessen (2018) for Epitope Research

1. Background & Rationale

Jessen (2018) demonstrated that a feed-forward neural network (FFN) could achieve ~95% accuracy classifying 9-mer peptides as Strong Binder (SB), Weak Binder (WB), or Non-Binder (NB) to HLA-A*02:01, using netMHCpan-4.0 labels as ground truth. This study reproduces that pipeline and extends it with:

- 5-fold cross-validation (referenced in Jessen’s conclusion as standard practice)
- Ensemble prediction (wisdom-of-the-crowd)
- Multiple encoding schemes (BLOSUM62 + one-hot + physicochemical properties)
- Performance comparison across FFN, CNN, LSTM, and Random Forest
- Generalization testing to additional HLA alleles

2. Research Questions

RQ	Question	Metric
RQ1	Can we reproduce Jessen’s FFN ~95% accuracy on HLA-A*02:01?	Accuracy, F1, AUC
RQ2	Does 5-fold CV ensemble outperform single train/test split?	Accuracy, F1
RQ3	Which encoding (BLOSUM62 vs one-hot vs AAindex) performs best?	Accuracy across encodings
RQ4	Does LSTM architecture improve over FFN for sequential peptide data?	Accuracy, F1
RQ5	How well does the model generalize to other common HLA alleles (A*01:01, B*07:02)?	Accuracy, F1

3. Methods Summary (adapted from Jessen 2018)

3.1 Data Generation

- Generate N random 9-mer peptides ($N = 100,0001,000,000$)
- Submit to netMHCpan-4.1 for binding prediction against target HLA alleles

- Labels: SB (%Rank < 0.5), WB (%Rank 0.52.0), NB (%Rank > 2.0)
- Balance by downsampling to equal class sizes
- Train/test split: 90/10 (RQ1) + 5-fold CV (RQ2)

3.2 Peptide Encoding

- **BLOSUM62**: 920 matrix per peptide (original method)
- **One-hot**: 920 binary matrix
- **AAindex**: 9N physicochemical property matrix (optional extension)

3.3 Model Architectures

Model	Architecture	Parameters
FFN	Dense(180)Dropout(0.4)Dense(90)Dropout(0.3)Dense(3, softmax)	~40K
CNN	Conv2D(32, 33)Dropout(0.25)FlattenFFN	~82K
ResNet-Style	Conv(32)BNReLU Residual Block3 (3264128 filters, skip connections + 11 projections) GlobalAvgPool Dense(128)Dropout Softmax	~150K
LSTM	LSTM(64)Dropout(0.3)Dense(32)Dense(3, softmax)	~45K
RF	100 trees, default mtry	~100 trees

3.4 Training

- Loss: categorical cross-entropy
- Optimizer: RMSprop (original) + Adam (extension)
- Epochs: 150, batch size: 50
- Validation split: 0.2 (internal holdout)
- Early stopping with patience = 10 (extension)

3.5 Evaluation

- Primary: Accuracy on held-out 10% test set
- Secondary: F1-score (per class + macro), AUROC (one-vs-rest)
- Confusion matrix visualization
- Per-class precision/recall

4. R Implementation Plan

The companion script `peptide_mhc_binding_study.R` implements:

1. **Setup & Dependencies** keras3/tensorflow, tidyverse, PepTools, random-Forest
2. **Data Module** peptide generation with unique `peptide_id`, netMHCpan prediction (or pre-computed data load), balancing
3. **Encoding Module** BLOSUM62, one-hot encodings
4. **Model Module** FFN, CNN, ResNet-Style, LSTM, Deep FFN, RF training functions
5. **Evaluation Module** metrics, confusion matrices, ROC curves
6. **Feature Extraction** raw BLOSUM features + penultimate-layer learned features `feature.csv`
7. **Model Persistence** all models saved to `models/*.h5` + `models/model_manifest.csv`
8. **Extension Module** 5-fold CV, ensemble, multi-allele testing

5. Expected Outcomes

- Reproduction of ~95% FFN accuracy on HLA-A*02:01 (RQ1)
- CV ensemble > single split performance (RQ2)
- BLOSUM62 > one-hot encoding (RQ3)
- LSTM FFN (sequential context helps marginally) (RQ4)
- Performance drop on alternate alleles proportional to training data availability (RQ5)

6. References

- Jessen, L.E. (2018). “Deep Learning for Cancer Immunotherapy.” RStudio AI Blog.
- Reynisson, B. et al. (2020). “NetMHCpan-4.1 and NetMHCIIpan-4.0.” *Nucleic Acids Research*.
- Oluwagbemi et al. (2025). “AI-driven epitope prediction: a systematic review.” *npj Vaccines*.